

MUHAMMAD ALIYAN



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PROFESSIONAL SUMMARY

Full-Stack MERN Developer with expertise in building scalable, production-ready web applications. Proven track record in developing end-to-end solutions with modern tech stacks, cloud deployment, and real-time systems. Strong foundation in AI/ML with hands-on experience in computer vision and deep learning. Passionate about clean architecture, performance optimization, and delivering high-impact software solutions.

EDUCATION

National University of Sciences and Technology (NUST), CEME <i>Bachelor of Engineering in Computer Engineering</i>	Rawalpindi, Pakistan <i>Graduating in May 2026</i>
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PROFESSIONAL EXPERIENCE

Full Stack Developer <i>Zapply (USA) – Remote</i>	Nov 2025 – Present <i>Full-Time</i>
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- Developing and maintaining scalable MERN-based applications for production environments.
- Collaborating with cross-functional teams on system design and feature delivery.

MERN Full Stack Developer <i>3Dim (China) – Remote</i>	Aug 2025 – Nov 2025 <i>Full-Time</i>
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- Built and deployed full-stack features using React.js, Node.js, and MongoDB.
- Improved API performance and frontend responsiveness.

Freelance Full Stack Developer <i>Fiverr – Remote</i>	Jun 2023 – Present <i>Self-Employed</i>
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- Delivered end-to-end web solutions for international clients, including dashboards, APIs, and authentication systems.

MERN Full Stack Developer Intern <i>Zapply (USA) – Remote</i>	Jun 2025 – Aug 2025 <i>Internship</i>
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- Assisted in building production-ready MERN features.

AI/ML Intern <i>CARE</i>	May 2025 – Jul 2025 <i>Internship</i>
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- Worked on machine learning and computer vision tasks involving classification and model evaluation.

LEADERSHIP & EXTRACURRICULAR

Director, Web & IT Wing <i>COMPPEC (NUST Computer Engineering Society)</i>	Jan 2025 – May 2025
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Deputy Director, Admin & HR <i>COMPPEC (NUST Computer Engineering Society)</i>	Jan 2024 – May 2024
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PROJECTS

Workify – Full-Stack Freelancing Platform <i>React, TypeScript, Node.js, MongoDB, Redis</i>	<u>Live</u>
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- Built freelancing marketplace enabling buyers and sellers to connect, post gigs, and manage orders.
- Designed scalable cloud architecture using Vercel, Render, MongoDB Atlas, Redis Cloud, and Azure Blob Storage.
- Implemented JWT authentication with Redux Toolkit and secure media uploads using Azure SAS tokens.
- Built dashboards with real-time analytics and engineered RESTful APIs with Mongoose indexing and Redis caching.

Full-Stack Social Media Application <i>MERN Stack, GridFS, Multer</i>
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- Developed social platform with posts, likes, comments, reels, and stories.
- Implemented bcrypt password hashing, JWT authentication, and MongoDB GridFS for media storage.

- Optimized performance through state management, server-side pagination, and media streaming.

Real-Time Chat Application | *MERN Stack, Socket.io*

- Built instant messaging with real-time communication, online/offline status, and media sharing.
- Implemented MongoDB schemas with GridFS and RESTful APIs for messaging operations.

AI-Powered Document Assistant "MERNBot" | *MERN Stack, OpenAI API*

- Integrated OpenAI GPT-3.5 for document summarization and AI-powered Q&A chatbot.
- Built RESTful APIs for text preprocessing and prompt engineering with JWT authentication.

Autonomous Vehicle System | *Raspberry Pi, OpenCV, Python*

- Designed self-driving car prototype using classical CV algorithms for real-time obstacle recognition.
- Implemented edge detection, contour analysis, and GPIO control for vehicle motion.
- Achieved obstacle avoidance and lane-following using classical CV on low-power hardware.

Mission Logging System for Pakistan Army | *Flutter, Django, SQL*

- Developed cross-platform mobile app for mission logging and tracking using Flutter and Dart.
- Built Django backend with RESTful APIs and implemented encrypted transmissions with RBAC.

Anti-Sleep Detection System | *Python, C++, Arduino, ESP32*

- Designed drowsiness detection system with eye closure analysis and automatic vehicle stopping.
- Created mobile app for real-time driver monitoring and remote vehicle control.

Audio Classification using CNN | *TensorFlow, Keras, Librosa*

- Built deep learning model for sound classification using UrbanSound8K dataset with 87% accuracy.
- Extracted MFCC features and implemented data augmentation techniques.

Hand & Face Detection | *OpenCV, MediaPipe, Python*

- Developed real-time detection system with hand landmark detection for gesture-based interaction.

TECHNICAL SKILLS

Web & Full-Stack: MERN Stack, RESTful APIs, JWT, Redux Toolkit, TypeScript, Socket.io, Tailwind CSS

Backend & APIs: Node.js, Express.js, Django, Django REST Framework, Flask, Multer, bcrypt

Databases: MongoDB, MySQL, PostgreSQL, Redis, SupaBase, GridFS

Cloud & DevOps: Docker, GitHub Actions (CI/CD), Vercel, Render, Azure Blob Storage

Machine Learning & AI: TensorFlow, Keras, Scikit-Learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn

Mobile: Flutter, Dart

Programming: Python, JavaScript, TypeScript, C++, Java

Tools: Git, GitHub, MLflow, DVC, Raspberry Pi, Arduino, OpenAI API

CERTIFICATIONS

- **Machine Learning Specialization** – Coursera (Andrew Ng)
- **Deep Learning Specialization** – Coursera (Andrew Ng)
- **Computer Vision and Image Processing** – Coursera
- **Advanced Computer Vision with TensorFlow** – Coursera
- **Supervised Machine Learning: Regression & Classification** – Coursera
- **Unsupervised Learning: Recommenders & Reinforcement Learning** – Coursera
- **Advanced Learning Algorithms** – Coursera
- **OpenCV Course** – freeCodeCamp